CS2201 Computer Organization & Architecture

L31: Tiled Chip Multicore Processors



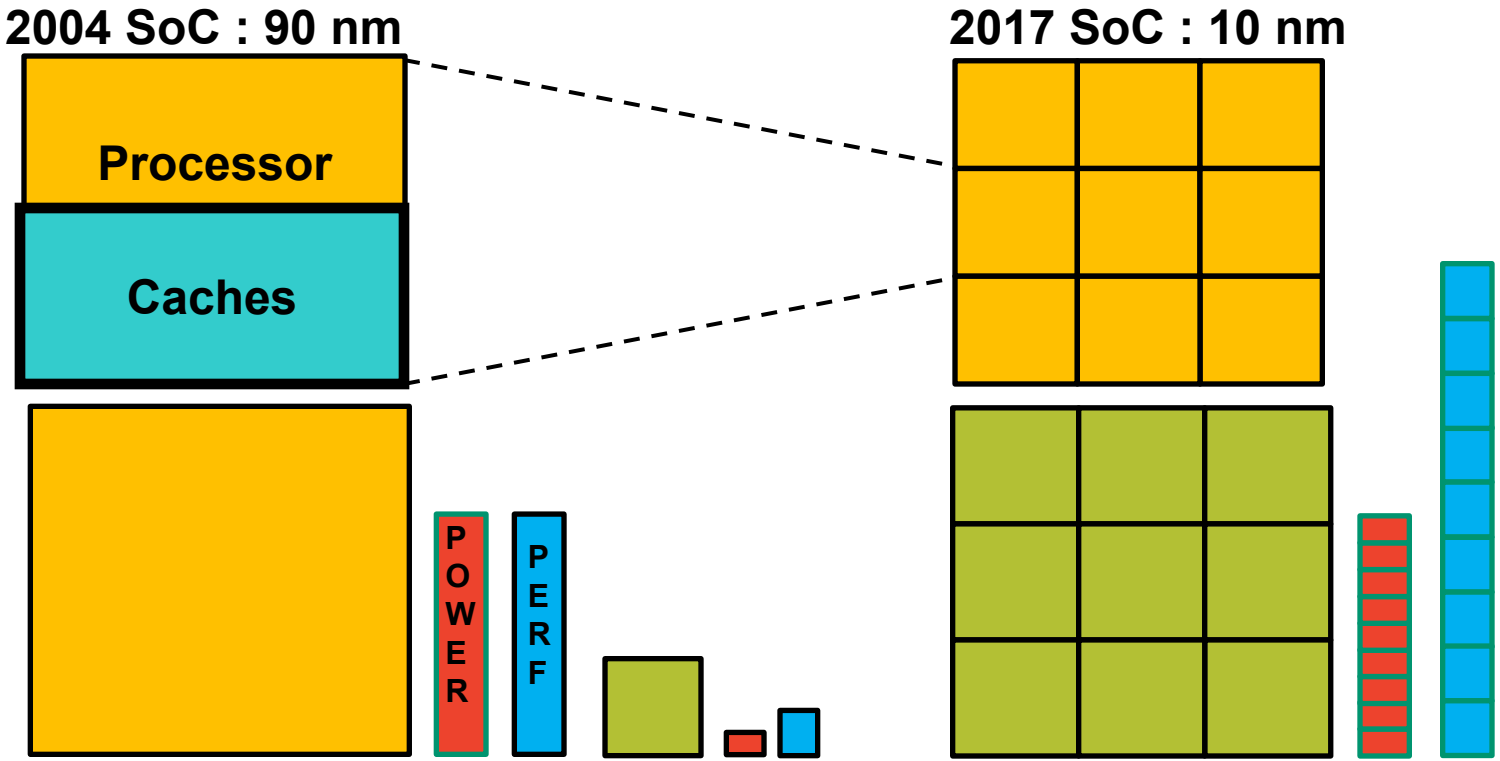
John Jose

Professor

Department of Computer Science & Engineering

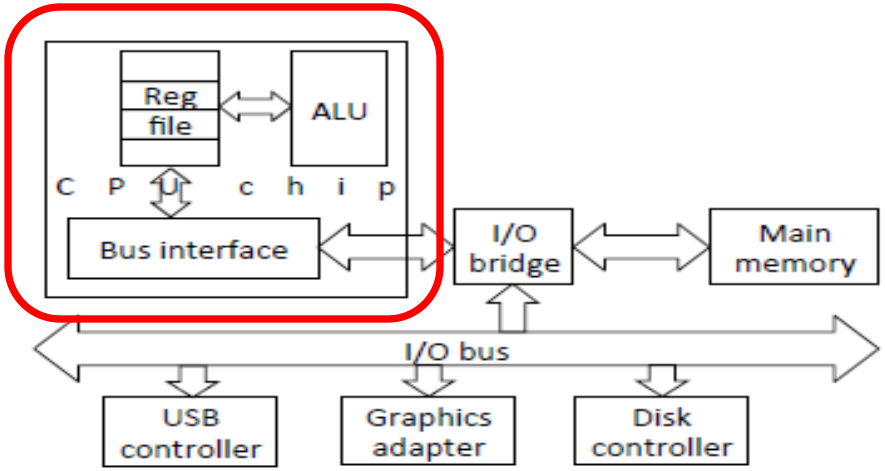
Indian Institute of Technology Guwahati

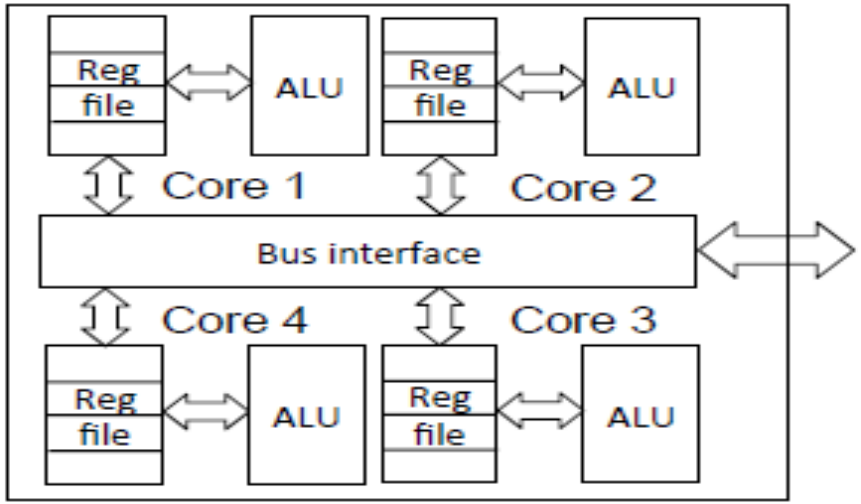
Paradigm Shift to Multicore



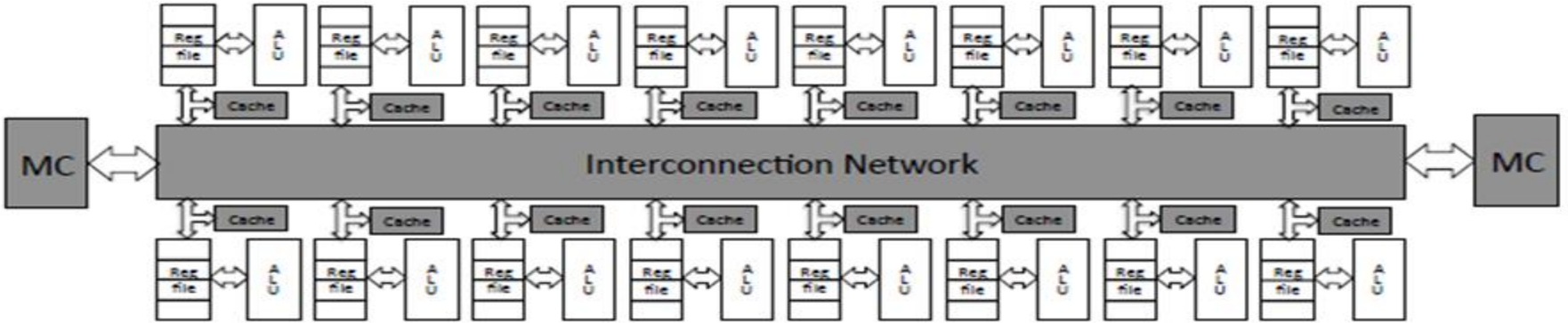
Multiple slower processors vs single fast powerful processor

What is Multicore ?



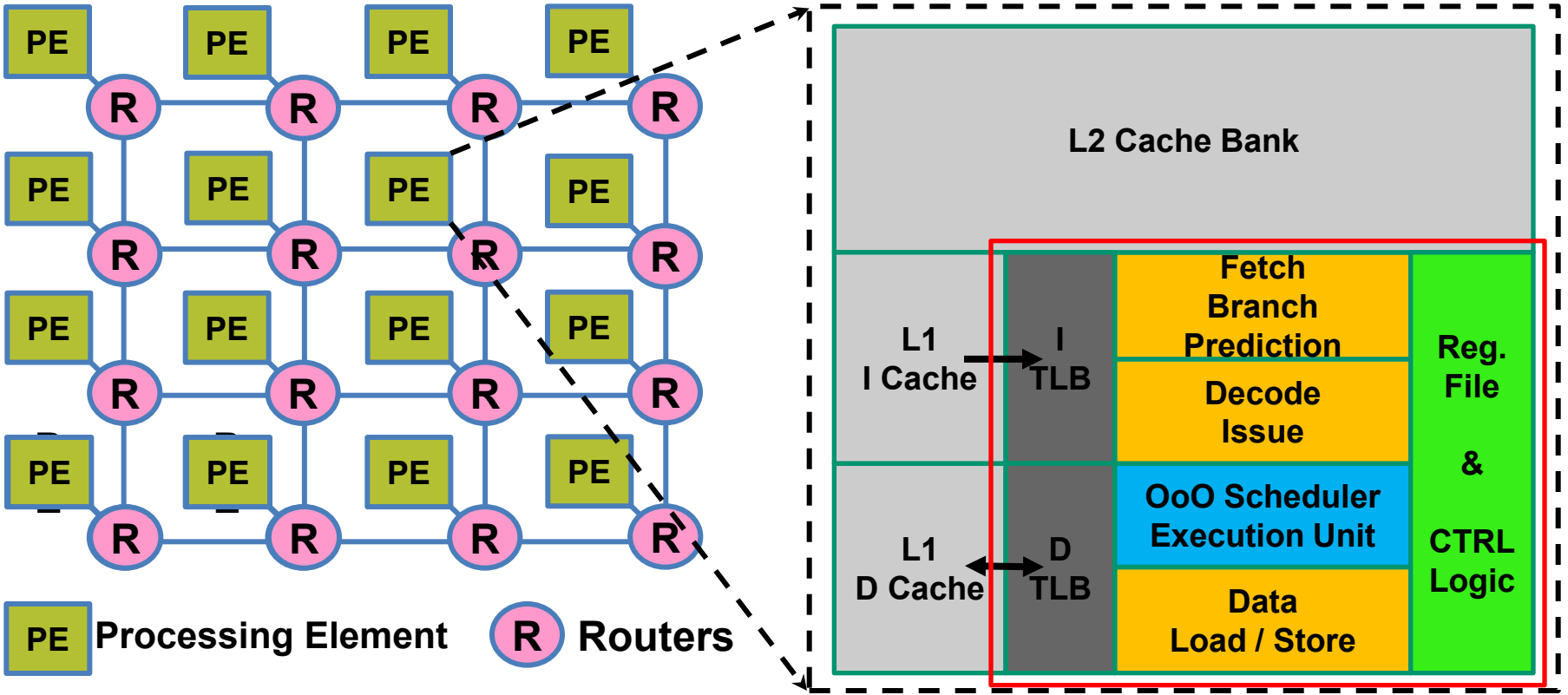


What is Multicore ?

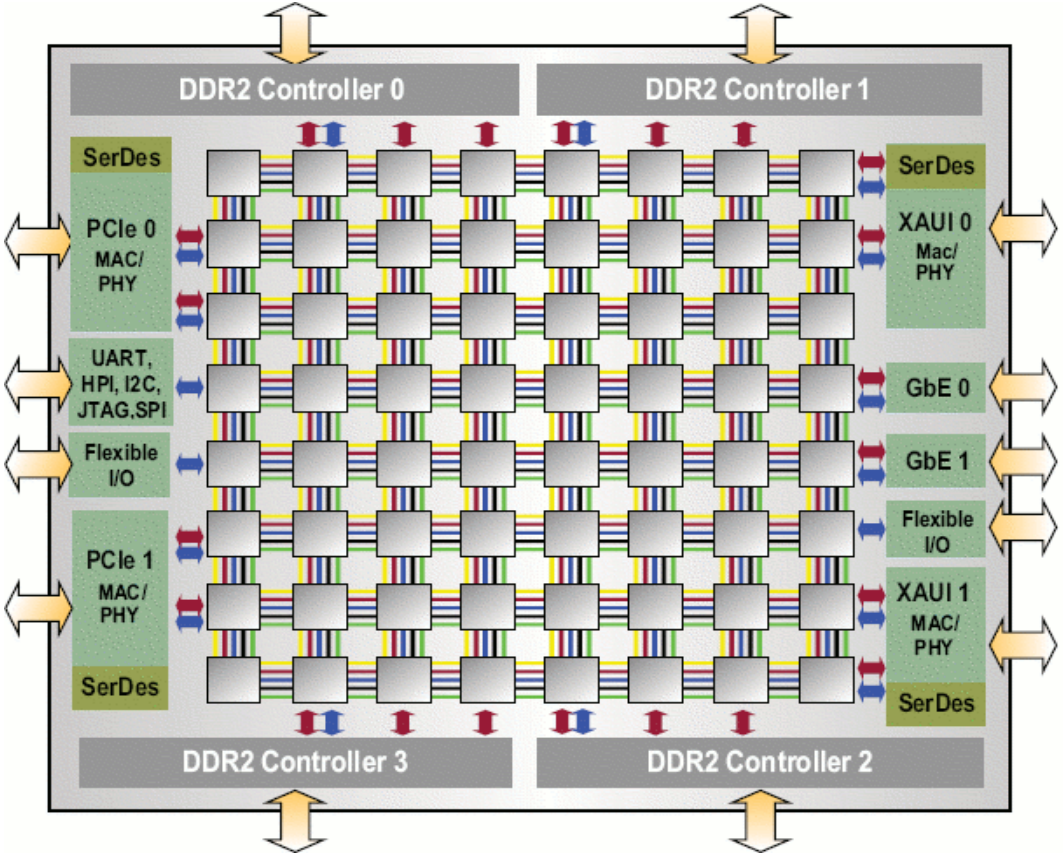


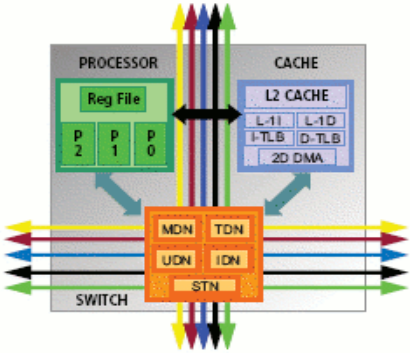
- How these cores communicate ?
- What is the best interconnection mechanism ?

Tiled Chip Many-Core Processor (TCMP)



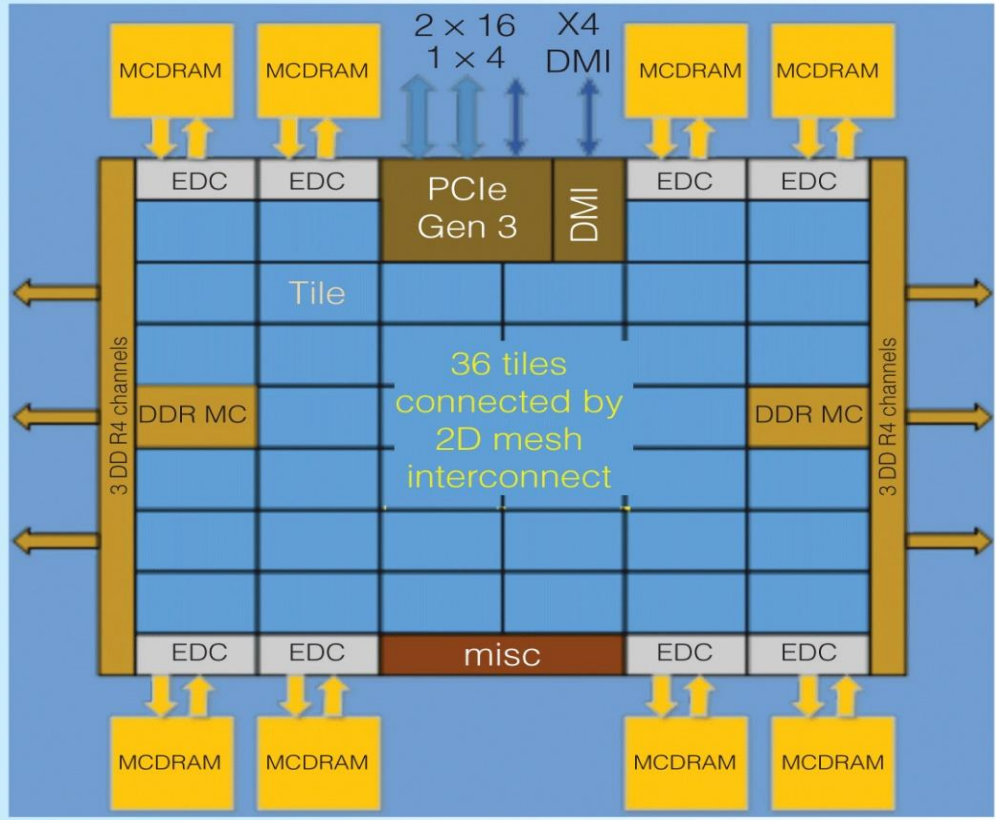
State-of-the-Art Architectures

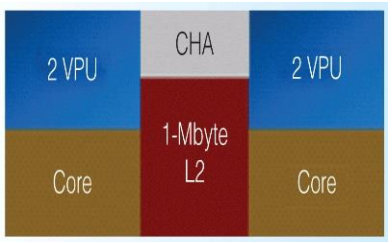




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State-of-the-Art Architectures

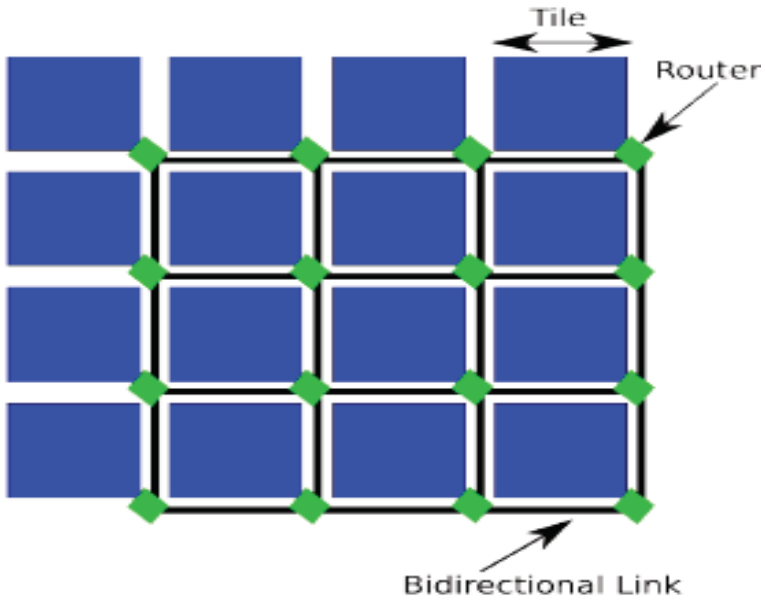




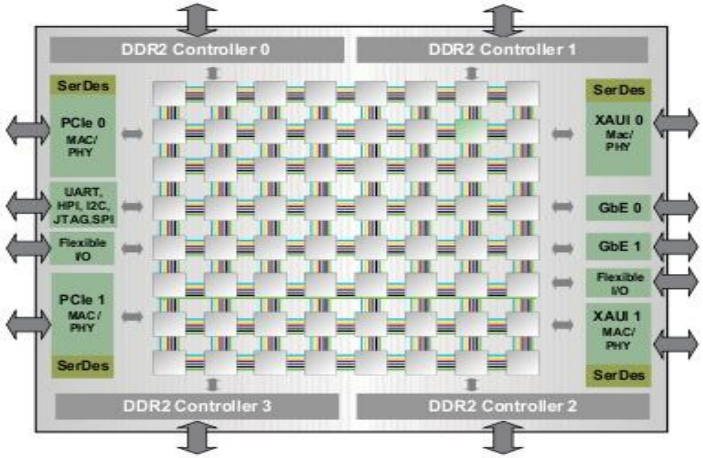
Intel KNL

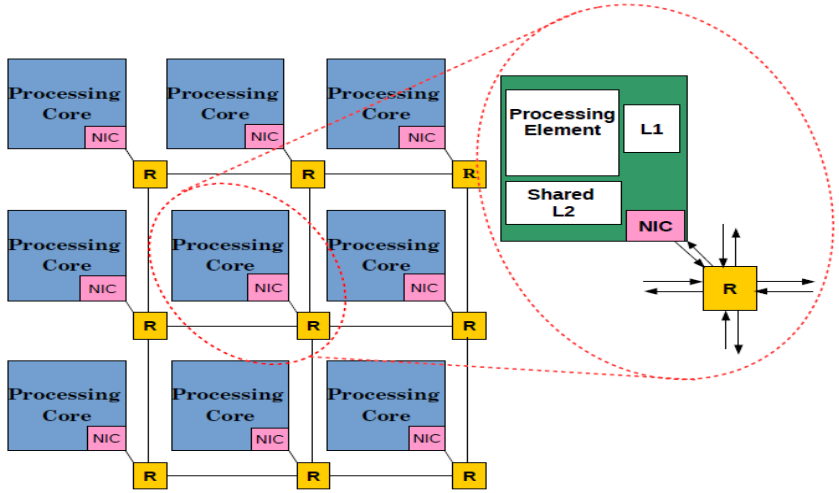
What is NoC ?

- Processing units interconnected via packet based network
- Each resource is called as a ***'tile'***
- All resources organized as rectangular tiles on the chip.
- Each tile have an address **- *(X,Y)***
- Tiles interconnected by network of routers
- Communication by packet transmission



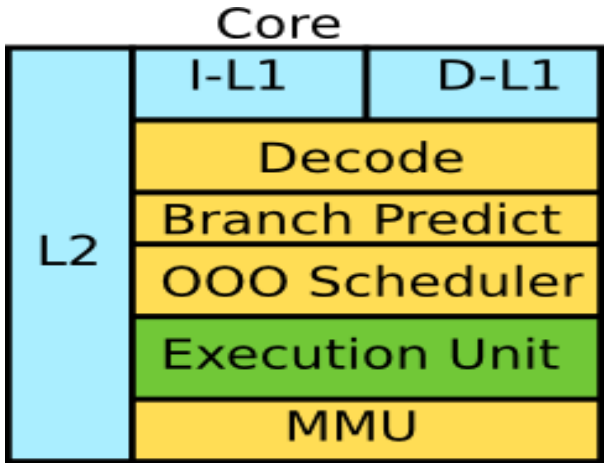
Routers and Tiles





- East, West, North and South neighbors
- Packets are divided into flow control units called flits
- L1 and L2 cache misses create NoC traffic packets

What is the role of on-chip cache ?

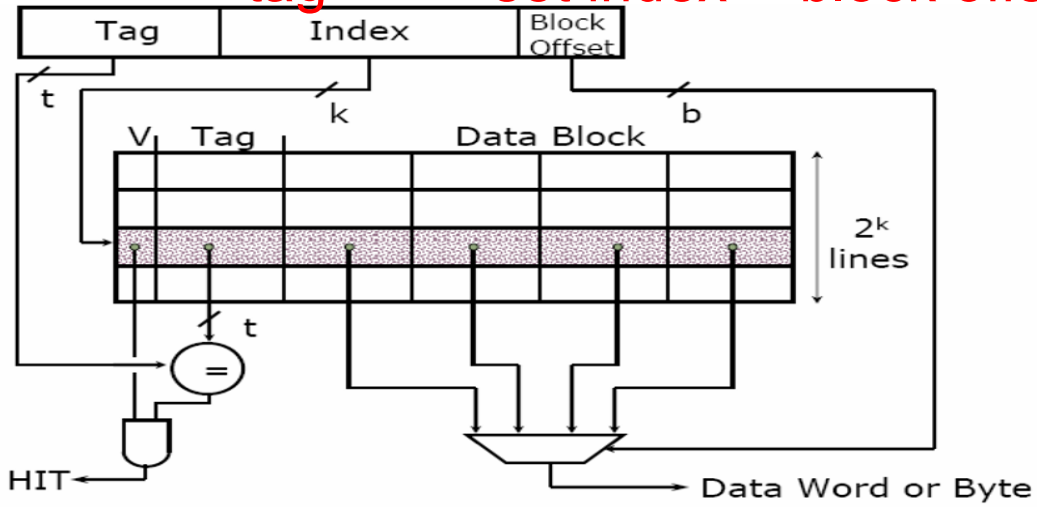


Address A: 1001 0011 0101 1010

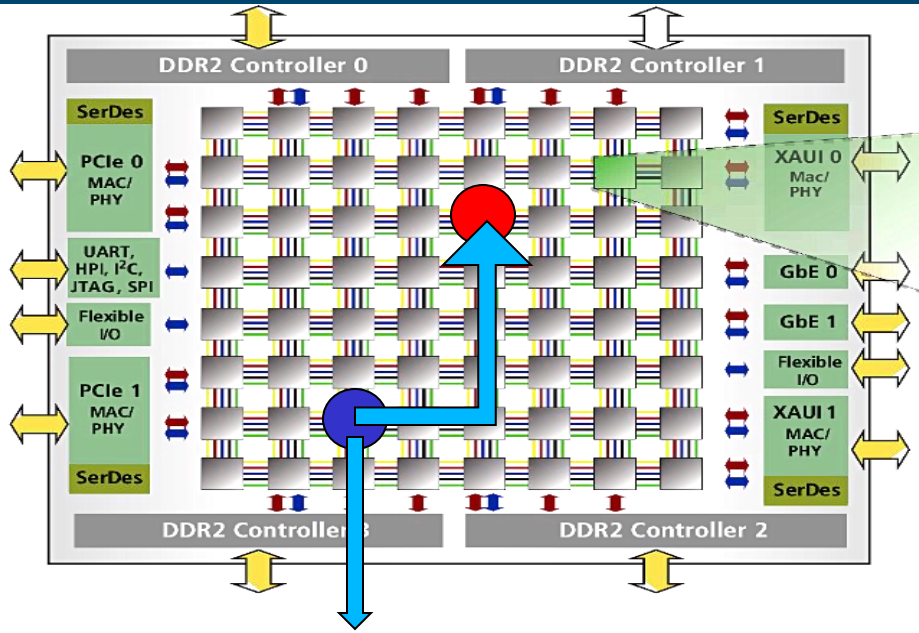
t bits s bits b bits

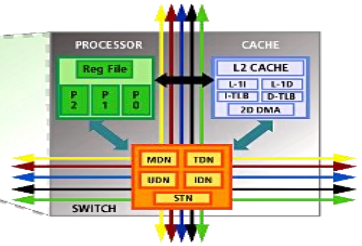
m-1 0

<tag> <set index> <block offset>



On-Chip Cache Address Mapping





Per core: 512 KB

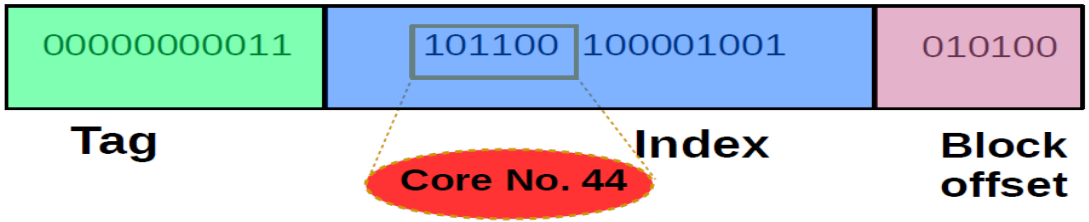
16-way set associative mapping

64 Bytes/block

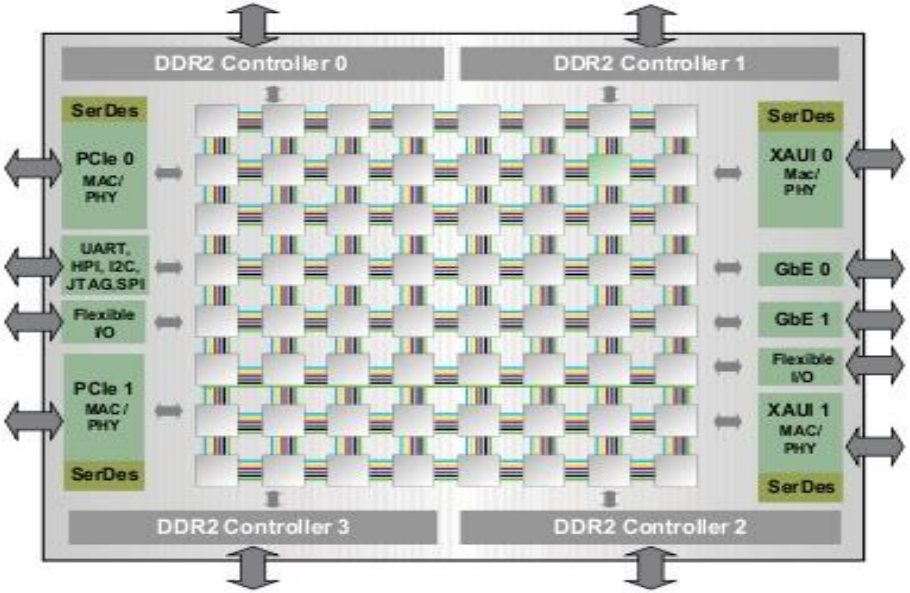
Consider a 4GB RAM (2^{32}bits)

Hex Address : 0x764254

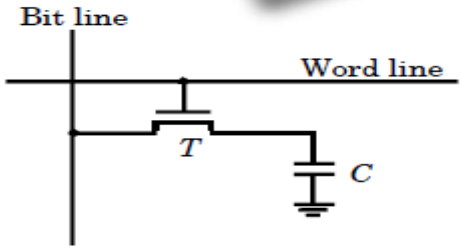
11 bits	15 bits	6 bits
Tag	Index	Block offset



What is the role of memory controllers?





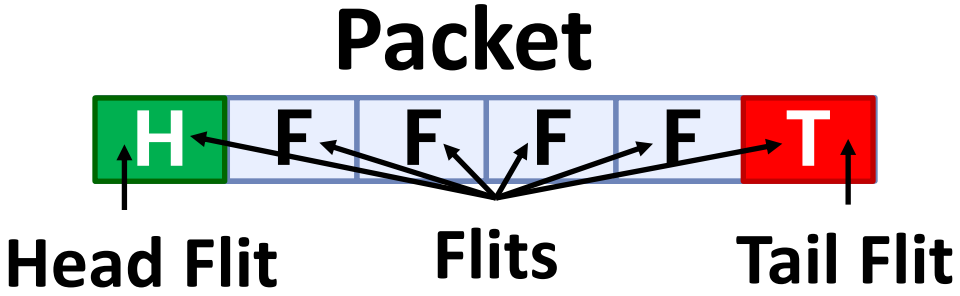


A single-transistor dynamic memory cell



Packets & Flits

- **Packet**
  - Unit of transfer for network
- **Flit**
  - Basic unit of transfer between a pair of routers
  - Unit of flow control within network





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